DECODO Unblocker Integration Guide

Overview

This guide describes the integration of DECODO Unblocker proxy service with the Eiros Agent iOS application for enhanced web automation and browsing capabilities.

Features

✅ Implemented Features

1. **ProxyManager** - Central proxy management system
   - DECODO primary and backup endpoints
   - SOCKS5 support for advanced tunneling
   - Automatic failover between proxies
   - Real-time latency monitoring
   - Connection health checks

2. **WebView Integration** - Seamless browser proxy support
   - Dynamic proxy configuration
   - Real-time proxy switching
   - Anti-detection headers
   - User agent rotation

3. **Automation Integration** - Proxy support for AI automation
   - Qwen2VL model requests through proxy
   - Automated web interactions via proxy
   - Proxy status monitoring in automation panel

4. **User Interface** - Complete proxy control interface
   - Proxy status indicators
   - Quick proxy toggle buttons
   - Detailed proxy configuration
   - Statistics and monitoring

Architecture

Core Components

```
ProxyManager (Singleton)
├── DECODO Configuration Management
├── Connection Health Monitoring
├── Automatic Failover Logic
└── Session Management

WebViewManager
├── Dynamic Proxy Configuration
├── WKWebView Proxy Integration
└── Real-time Proxy Switching

AutomationController
├── Proxy-enabled Automation
├── Qwen2VL Proxy Integration
└── Status Monitoring

UI Components
├── ProxyControlView (Full Control Panel)
├── BrowserView (Proxy Status Indicator)
└── AutomationPanelView (Quick Controls)
```

DECODO Endpoints

- **Primary HTTP**: `proxy.decodo.io:8080`
- **Backup HTTP**: `backup.decodo.io:8080`
- **SOCKS5**: `proxy.decodo.io:1080`

Usage

1. Basic Proxy Control

```swift
// Enable DECODO primary proxy
await ProxyManager.shared.enableDECODOPrimary()

// Check proxy status
let isActive = ProxyManager.shared.isUsingProxy()

// Get proxy information
let info = ProxyManager.shared.getCurrentProxyInfo()
```

2. WebView with Proxy

```swift
// WebView automatically uses proxy when enabled
let webViewManager = WebViewManager()

// Check proxy status in WebView
let status = webViewManager.getDECODOStatus()

// Enable proxy for WebView
await webViewManager.enableDECODOProxy()
```

3. Automation with Proxy

```
// Enable proxy for automation
await AutomationController.shared.enableDECODOForAutomation()

// Run automation task with proxy
let task = AutomationTask(...)
await AutomationController.shared.executeTask(task)
```

4. UI Integration

The app includes several UI components for proxy control:

- **Browser View**: Proxy status indicator with quick access
- **Automation Panel**: Proxy toggle and status display
- **Proxy Control View**: Full configuration and monitoring

Configuration

Default Settings

```
// Connection settings
static let connectionTimeout: TimeInterval = 30.0
static let maxRetries = 3
static let healthCheckInterval: TimeInterval = 60.0

// Anti-detection features
static let userAgentRotation = true
static let headerRandomization = true
static let connectionPooling = true
```

Custom Configuration

You can add custom proxy configurations through the UI or programmatically:

```
let customProxy = ProxyConfiguration(
    id: "custom-proxy",
    name: "Custom Proxy",
    type: .http,
    host: "your-proxy.com",
    port: 8080,
    username: "user",
    password: "pass",
    isDefault: false,
    capabilities: [.webUnblocking]
)

ProxyManager.shared.saveCustomProxyConfiguration(customProxy)
```

Anti-Detection Features

1. User Agent Rotation

- Pool of realistic mobile and desktop user agents
- Automatic rotation on each request

- iOS Safari, Chrome, and Firefox variants

2. Header Randomization

- Realistic browser headers
- DNT (Do Not Track) support
- Proper Accept headers for different content types

3. Connection Management

- Connection pooling for efficiency
- Request throttling to avoid detection
- Proper SSL/TLS handling

Monitoring and Diagnostics

1. Real-time Monitoring

- Connection latency tracking
- Data transfer statistics
- Proxy health status
- Automatic failover events

2. Logging

- Detailed proxy connection logs
- Error tracking and reporting
- Performance metrics
- Debug information

3. Testing

- Built-in proxy connection testing
- Multiple test endpoints
- Latency measurement
- Success rate tracking

Troubleshooting

Common Issues

1. **Proxy Connection Failed**
   - Check internet connectivity
   - Verify DECODO service status
   - Try backup proxy endpoint

2. **High Latency**
   - Switch to backup proxy
   - Check network conditions
   - Verify proxy server load

3. **WebView Not Using Proxy**
   - Ensure proxy is enabled in ProxyManager

- Check WebView configuration
- Verify proxy settings are applied

Debug Commands

```
// Test all DECODO proxies
let results = await ProxyManager.shared.testAllDECODOProxies()

// Get detailed status
let status = AutomationController.shared.getAutomationStatus()

// Check component readiness
let components = status.componentsReady
```

Security Considerations

1. **No Authentication Required**: DECODO typically doesn't require username/password
2. **HTTPS Support**: All HTTPS traffic is properly tunneled
3. **DNS Resolution**: DNS queries go through proxy for privacy
4. **No Logging**: Local app doesn't log sensitive proxy traffic

Performance Optimization

1. **Connection Pooling**: Reuse connections for efficiency
2. **Automatic Failover**: Switch to backup proxy on failure
3. **Health Monitoring**: Proactive connection health checks
4. **Latency Optimization**: Choose fastest available proxy

Integration Testing

Test Scenarios

1. **Basic Connectivity**
   - Enable proxy and browse websites
   - Verify IP address changes
   - Test HTTPS sites

2. **Automation Testing**
   - Run automation tasks with proxy enabled
   - Verify Qwen2VL model requests work
   - Test web scraping scenarios

3. **Failover Testing**
   - Disable primary proxy
   - Verify automatic failover to backup
   - Test recovery when primary returns

4. **Performance Testing**
   - Measure latency with/without proxy
   - Test concurrent connections
   - Monitor memory usage

Future Enhancements

Planned Features

1. **Geographic Proxy Selection**
   - Choose proxy by country/region
   - Automatic geo-optimization

2. **Load Balancing**
   - Distribute requests across multiple proxies
   - Smart routing based on performance

3. **Advanced Analytics**
   - Detailed usage statistics
   - Performance trending
   - Cost optimization insights

4. **Custom Proxy Protocols**
   - Support for additional proxy types
   - Custom authentication methods
   - Protocol-specific optimizations

Support

For issues related to DECODO integration:

1. Check proxy service status at DECODO dashboard
2. Review application logs for error details
3. Test with backup proxy endpoints
4. Contact DECODO support for service issues

Changelog

Version 1.0.0 (2025-09-22)

- Initial DECODO Unblocker integration
- ProxyManager implementation
- WebView proxy support
- Automation proxy integration
- Complete UI for proxy control
- Anti-detection features
- Health monitoring and failover
- Documentation and testing guides